

# Emergency Mesh: Technical Reference

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BUILT

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COMMIT

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GENERATOR

docs/tools/build\_technical\_reference.py

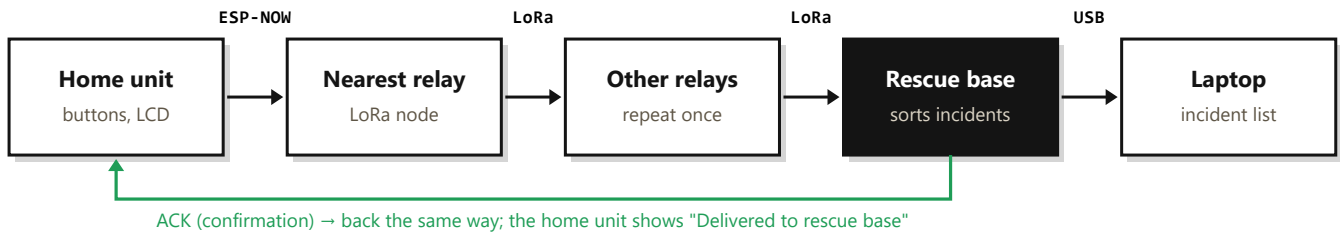
Every number in this document is read from the repository (code, results and README) or computed from it by the generator. Facts from outside sources are listed in section 10. Where text is quoted from the README, the section numbers in it are README sections.

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|   |                           |    |                      |
|---|---------------------------|----|----------------------|
| 2 | System at a glance        | 8  | Results              |
| 3 | Packet format             | 9  | Duty-cycle check     |
| 4 | Time on air               | 10 | Outside facts        |
| 5 | Radio settings            | 11 | Wokwi prototype pins |
| 6 | Node rules and forwarding | 12 | Known limits         |
| 7 | Simulator setup           |    |                      |

## 2 System at a glance

A person presses a button on their home unit. The SOS goes over ESP-NOW to the nearest relay node, then relay to relay over LoRa to the rescue base. The base confirms with an ACK that travels back the same way.



Path of one SOS: home unit → nearest relay → other relays → rescue base → laptop. LoRa between relays at SF9, 125 kHz (`RadioConfig`); every relay repeats each new message once.

### What is built and what is designed only

| Part                                                                                                                                                                 | Status                                                                                   | Where                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Mesh logic: message IDs, duplicate dropping, hop limits, priority queue, random delays, listen-before-talk, delivery confirmations, retries, rate limits, heartbeats | <b>Built and tested</b> in a Python network simulator (16 automated tests)               | <code>emergency-mesh/simulation/</code>                                |
| Disaster scenarios: rerouting, node failures, mass SOS, dead-node detection, sensitivity to radio assumptions                                                        | <b>Built</b> , results in Section 3                                                      | <code>emergency-mesh/simulation/results/</code>                        |
| Packet format (max 31 bytes) and LoRa time-on-air                                                                                                                    | <b>Built</b>                                                                             | <code>protocol.py</code>                                               |
| Node interface: OLED screen, SOS button, status LED                                                                                                                  | <b>Built</b> in the Wokwi ESP32 simulator (MicroPython)                                  | <code>emergency-mesh/wokwi_node/</code> , <a href="#">live project</a> |
| Interactive map demo: click anywhere to send an SOS, destroy or repair nodes, simulate a crowd                                                                       | <b>Built</b> , runs in any web browser                                                   | <code>emergency-mesh/web_demo/</code>                                  |
| Rescue dashboard                                                                                                                                                     | <b>Prototype</b> : priority-sorted incident table from the simulator (also saved as CSV) | <code>python run.py demo</code>                                        |
| ESP32 firmware with LoRa radio                                                                                                                                       | <b>Designed</b> , not built                                                              | Sections 5–6                                                           |
| Home unit (ESP-NOW)                                                                                                                                                  | <b>Designed</b> , not built                                                              | Sections 6 and 7                                                       |
| Schematics: system diagram, node block diagram, Wokwi circuit                                                                                                        | <b>Included</b>                                                                          | Sections 2 and 6                                                       |
| Physical nodes                                                                                                                                                       | <b>Not done yet</b>                                                                      | Section 10                                                             |
| Field range tests for LoRa and ESP-NOW                                                                                                                               | <b>Not done yet</b>                                                                      | Section 10                                                             |

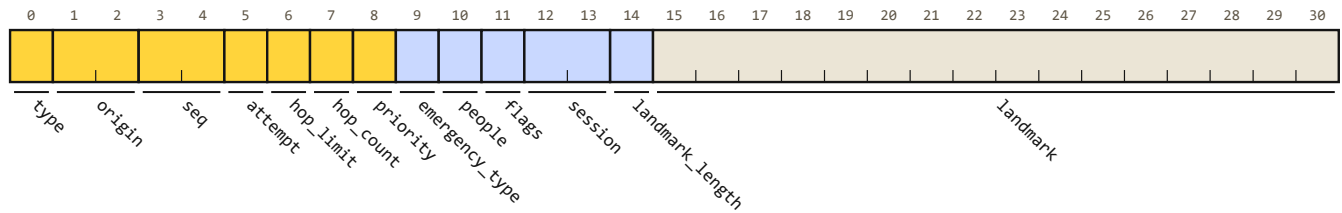
From the README, "Project status". Section numbers in this table are README sections.

## 3

## Packet format

All three packet types share a 9-byte header. Multi-byte fields are little-endian ( `struct` format prefix `<` ), so the low byte is sent first. Layouts are read from `protocol.py`.

### SOS packet, bytes 0–30 (15–31 bytes)

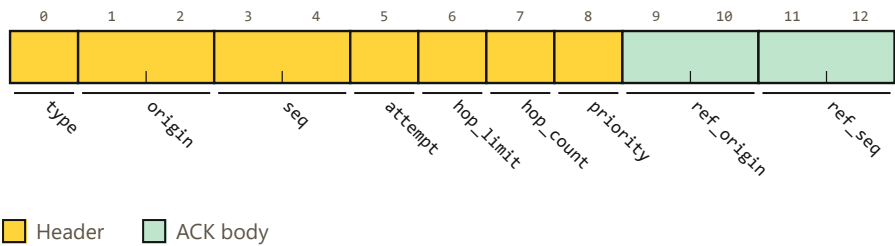


■ Header
 ■ SOS body
 ■ Landmark (optional)

| Bytes | Field           | Size   | Type                  | Meaning                                                                                                |
|-------|-----------------|--------|-----------------------|--------------------------------------------------------------------------------------------------------|
| 0     | type            | 1 B    | uint8                 | SOS, ACK or HEARTBEAT                                                                                  |
| 1–2   | origin          | 2 B    | uint16, little-endian | Node where the message was created                                                                     |
| 3–4   | seq             | 2 B    | uint16, little-endian | Counter on that node; (origin, seq) identifies the emergency                                           |
| 5     | attempt         | 1 B    | uint8                 | 1 for the first send, 2+ for retries                                                                   |
| 6     | hop_limit       | 1 B    | uint8                 | Starts at 6, drops by 1 per hop                                                                        |
| 7     | hop_count       | 1 B    | uint8                 | Hops travelled so far                                                                                  |
| 8     | priority        | 1 B    | uint8                 | 3 critical, 2 urgent, 1 supplies, 0 "I'm safe"                                                         |
| 9     | emergency_type  | 1 B    | uint8                 | Medical, trapped, flooding, fire, collapse, other                                                      |
| 10    | people          | 1 B    | uint8                 | Number of people                                                                                       |
| 11    | flags           | 1 B    | uint8                 | Injured, trapped, child, elderly, water rising                                                         |
| 12–13 | session         | 2 B    | uint16, little-endian | Sender ID: the home unit's ID, used for rate limiting and to look up the address registered at install |
| 14    | landmark_length | 1 B    | uint8                 | Number of landmark bytes that follow                                                                   |
| 15–30 | landmark        | 0–16 B | UTF-8 text            | e.g. "Blue gate, Ln 3" (plus 1 length byte)                                                            |

Header `struct.Struct("<BHHBBBB")` = 9 B; SOS body `struct.Struct("<BBBHB")` = 6 B; landmark  $\leq$  16 B (`MAX_LANDMARK_BYTES`), cut so a UTF-8 character is never split. Meanings from README section 5.

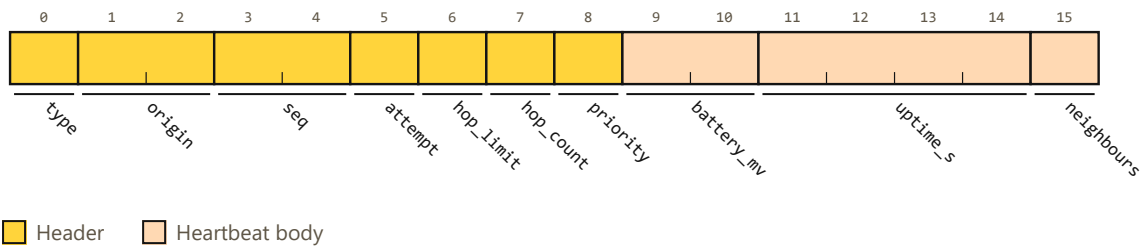
ACK packet (13 bytes)



| Bytes | Field      | Size | Type                  | Meaning                           |
|-------|------------|------|-----------------------|-----------------------------------|
| 9–10  | ref_origin | 2 B  | uint16, little-endian | origin of the SOS being confirmed |
| 11–12 | ref_seq    | 2 B  | uint16, little-endian | seq of the SOS being confirmed    |

Bytes 0–8 are the same header as the SOS packet.

Heartbeat packet (16 bytes)



| Bytes | Field      | Size | Type                  | Meaning                        |
|-------|------------|------|-----------------------|--------------------------------|
| 9–10  | battery_mv | 2 B  | uint16, little-endian | Battery voltage in millivolts  |
| 11–14 | uptime_s   | 4 B  | uint32, little-endian | Seconds since the node started |
| 15    | neighbours | 1 B  | uint8                 | Nodes within radio range       |

Codes

| Code | type      |
|------|-----------|
| 1    | SOS       |
| 2    | ACK       |
| 3    | HEARTBEAT |

| Code | emergency_type |
|------|----------------|
| 1    | MEDICAL        |
| 2    | TRAPPED        |
| 3    | FLOODING       |
| 4    | FIRE           |
| 5    | COLLAPSE       |
| 6    | OTHER          |

| Code | priority | Meaning                                     |
|------|----------|---------------------------------------------|
| 3    | CRITICAL | life-threatening: injured, trapped, medical |
| 2    | URGENT   | water rising, children or elderly at risk   |
| 1    | SUPPLIES | food, water, medicine needed                |
| 0    | SAFE     | "I'm safe" check-in                         |

| Value | Bit   | flags        |
|-------|-------|--------------|
| 1     | bit 0 | INJURED      |
| 2     | bit 1 | TRAPPED      |
| 4     | bit 2 | CHILD        |
| 8     | bit 3 | ELDERLY      |
| 16    | bit 4 | WATER_RISING |

Flags combine by adding bits: INJURED|TRAPPED = 3.

## Worked example: `encode()` of the first SOS in the rerouting demo

The critical SOS from node 1 in `run.py demo` (seed 7): trapped, 4 people, flags INJURED|TRAPPED, landmark "School rooftop". It encodes to 29 bytes:

```
01 01 00 01 00 01 06 00 03 02 04 03 65 00 0e 53 63 68 6f 6f 6c 20 72 6f 6f 66 74 6f 70
```

| Bytes | Field           | Hex                                       | Value               |
|-------|-----------------|-------------------------------------------|---------------------|
| 0     | type            | 01                                        | 1 (SOS)             |
| 1-2   | origin          | 01 00                                     | 1                   |
| 3-4   | seq             | 01 00                                     | 1                   |
| 5     | attempt         | 01                                        | 1                   |
| 6     | hop_limit       | 06                                        | 6                   |
| 7     | hop_count       | 00                                        | 0                   |
| 8     | priority        | 03                                        | 3 (CRITICAL)        |
| 9     | emergency_type  | 02                                        | 2 (TRAPPED)         |
| 10    | people          | 04                                        | 4                   |
| 11    | flags           | 03                                        | 3 (INJURED TRAPPED) |
| 12-13 | session         | 65 00                                     | 101                 |
| 14    | landmark_length | 0e                                        | 14                  |
| 15-28 | landmark        | 53 63 68 6f 6f 6c 20 72 6f 6f 66 74 6f 70 | "School rooftop"    |

Built by calling `encode()` on the packet the simulator created; `decode()` gives back the same packet. It matches row 1 of `results/demo_dashboard.csv`.

## 4 Time on air

Each packet's time on air comes from `lora_airtime()` in `protocol.py`, the SX127x formula published by Semtech. Settings: SF9, 125 kHz, coding rate 4/5, 8-symbol preamble, CRC on, explicit header.

### Our packets at SF9

| Packet                | Size | Payload symbols | Time on air |
|-----------------------|------|-----------------|-------------|
| ACK                   | 13 B | 28              | 164.9 ms    |
| SOS, no landmark      | 15 B | 28              | 164.9 ms    |
| Heartbeat             | 16 B | 28              | 164.9 ms    |
| SOS, longest landmark | 31 B | 48              | 246.8 ms    |

### The formula, as coded

```
t_sym = (2 ** spreading_factor) / bandwidth_hz
low_dr = 1 if t_sym > 0.016 else 0
ih = 0 if explicit_header else 1
numerator = 8 * payload_bytes - 4 * spreading_factor + 28 + 16 * int(crc) - 20 * ih
payload_symbols = 8 + max(
    math.ceil(numerator / (4 * (spreading_factor - 2 * low_dr))) * (coding_rate + 4), 0)
return (preamble_symbols + 4.25) * t_sym + payload_symbols * t_sym
```

### Worked example: a 31-byte SOS at SF9

| Step                       | Calculation                                                            | Result          |
|----------------------------|------------------------------------------------------------------------|-----------------|
| Symbol time                | $2^9 / 125,000 \text{ Hz}$                                             | 4.096 ms        |
| Low data rate optimisation | on only if a symbol is longer than 16 ms                               | off             |
| Numerator                  | $8 \times 31 - 4 \times 9 + 28 + 16 \times 1 - 20 \times 0$            | 256             |
| Blocks                     | $\lceil 256 / (4 \times (9 - 2 \times 0)) \rceil = \lceil 7.11 \rceil$ | 8               |
| Payload symbols            | $8 + 8 \times (1 + 4)$                                                 | 48              |
| Preamble symbols           | $8 + 4.25$                                                             | 12.25           |
| Time on air                | $(12.25 + 48) \times 4.096 \text{ ms}$                                 | <b>246.8 ms</b> |

### Why several sizes take the same time

The payload is sent in blocks: each block carries 36 bits ( $4 \times \text{SF9}$ ) and costs 5 symbols, and a part-filled block costs the same as a full one. So time on air rises in steps:

| Packet size | Payload symbols | Time on air |
|-------------|-----------------|-------------|
| 13–17 B     | 28              | 164.9 ms    |
| 18–21 B     | 33              | 185.3 ms    |
| 22–26 B     | 38              | 205.8 ms    |
| 27–30 B     | 43              | 226.3 ms    |
| 31 B        | 48              | 246.8 ms    |

The ACK (13 B) and heartbeat (16 B) fall in the same step as an SOS with no landmark (15 B).

## Spreading factor

| Spreading factor | Symbol time | Low data rate opt. | ACK (13 B) | SOS (31 B) |
|------------------|-------------|--------------------|------------|------------|
| SF7              | 1.024 ms    | off                | 46.3 ms    | 71.9 ms    |
| SF8              | 2.048 ms    | off                | 82.4 ms    | 133.6 ms   |
| SF9              | 4.096 ms    | off                | 164.9 ms   | 246.8 ms   |
| SF10             | 8.192 ms    | off                | 288.8 ms   | 452.6 ms   |
| SF11             | 16.384 ms   | on                 | 577.5 ms   | 905.2 ms   |
| SF12             | 32.768 ms   | on                 | 1155.1 ms  | 1810.4 ms  |

SF7–SF12 is the range `run.py --sf` accepts; all results use SF9.

## 5 Radio settings

**Assumed, not measured.** No radio has been tested in the field yet. The range and loss figures come from published sources and our own allowances. README section 3.5 shows how much the results depend on them.

### In the code

| Setting            | Value     | Where                                           |
|--------------------|-----------|-------------------------------------------------|
| Spreading factor   | SF9       | <code>RadioConfig.spreading_factor</code>       |
| Bandwidth          | 125 kHz   | <code>RadioConfig.bandwidth_hz</code>           |
| Coding rate        | 4/5       | <code>lora_airtime(coding_rate=...)</code>      |
| Preamble           | 8 symbols | <code>lora_airtime(preamble_symbols=...)</code> |
| Range              | 800 m     | <code>RadioConfig.range_m</code>                |
| Random packet loss | 2%        | <code>RadioConfig.loss_probability</code>       |

## Why these values (README section 4)

| Value                                             | Reason                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 865–868 MHz band                                  | Licence-exempt in India under the Use of Low Power Equipment in the Frequency Band 865–868 MHz for Short Range Devices (Exemption from Licence) Rules, 2021 (G.S.R. 853(E)). For tracking, tracing and data acquisition devices: at most 500 mW e.r.p., channels up to 200 kHz, transmitting at most 2.5% of the time (10% for network access points), adaptive power control, and type-approved equipment |
| 866.0 MHz                                         | Inside that band, with room on both sides for a 125 kHz channel                                                                                                                                                                                                                                                                                                                                            |
| ESP-NOW on 2.4 GHz (home unit to nearest relay)   | Licence-exempt in India                                                                                                                                                                                                                                                                                                                                                                                    |
| SF9, 125 kHz, coding rate 4/5                     | SF9 is the suggested starting point for Indian outdoor projects; 125 kHz and 4/5 are the standard settings ( <a href="#">Zbotic SX1276 guide</a> )                                                                                                                                                                                                                                                         |
| 800 m range                                       | The same guide lists 500 m–1 km at SF9 in dense Indian cities (Delhi, Mumbai, Bangalore) for an SX1276 at +20 dBm with simple antennas. 800 m sits inside that range; Section 3.5 tests 500–1000 m                                                                                                                                                                                                         |
| +20 dBm transmit power                            | The SX1276's maximum (100 mW), which the range figure assumes; under the 500 mW e.r.p. limit with a simple antenna                                                                                                                                                                                                                                                                                         |
| 200 m home unit reach (interactive demo, assumed) | Our assumption for the demo. A published ESP-NOW test (Espressif developer blog, "ESP-NOW for outdoor applications", by a community author) found about 150 m reliable with built-in antennas, about 60% delivery at 300 m in an open field, and about 100% to 450 m in long-range mode. Not measured by us                                                                                                |
| 0.25 s per SOS on air                             | Calculated with the time-on-air formula in Semtech's SX1276 datasheet, for a 31-byte packet at SF9                                                                                                                                                                                                                                                                                                         |

## 6 Node rules and forwarding

Every `NodeConfig` field, its value in `node.py`, and the reason from README section 4. The generator checks that each value matches the README.

| In the code                                                                   | README                                           | Why                                                                                                                                                                                               |
|-------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hop_limit = 6</code>                                                    | Hop limit 6                                      | Farthest node is 4 hops away; 2 spare hops allow detours around destroyed nodes                                                                                                                   |
| <code>seen_cache_size = 256</code>                                            | Remember last 256 message IDs                    | Would use about 1.5 KB on the ESP32. A 16× larger memory didn't change the heavy-load results, so 256 is enough                                                                                   |
| <code>ack_timeout_s = 90</code><br><code>max_attempts = 5</code>              | Retry after 90 s, up to 5 sends                  | Normal delivery takes seconds (median 2.5–12.5 s in Section 3.2), so a retry means the message was truly lost. Waits double each time, so the last retry is about 23 minutes after the first send |
| <code>jitter_max_s = 1</code><br><code>busy_backoff_s = (0.2, 1.5)</code>     | Wait up to 1 s before sending; 0.2–1.5 s if busy | Spreads transmissions over several packet lengths (one SOS ≈ 0.25 s) to reduce collisions                                                                                                         |
| <code>rate_limit_count = 3</code><br><code>rate_limit_window_s = 600</code>   | 3 SOS per home unit per 10 minutes               | Lets someone update their SOS but stops one unit flooding the network                                                                                                                             |
| <code>heartbeat_interval_s = 900</code><br><code>silent_alert_s = 1800</code> | Heartbeat 15 min, alert at 30 min                | An alert needs two missed heartbeats, so one lost packet doesn't cause a false alarm                                                                                                              |

### Forwarding rules (README section 5)

1. Drop any message ID already seen.
2. Forward only while `hop_limit > 1`.
3. Always send the highest-priority waiting packet first.
4. Wait a random delay before sending; back off if the channel is busy.
5. Copies travel along every surviving path, so there is no route to repair when a node dies.
6. The base confirms every SOS attempt it receives.
7. Without a confirmation, the sender retries with doubling waits, up to 5 sends.
8. Rate limit per sender ID (one per home unit); heartbeats from every node.

## 7 Simulator setup

`grid_nodes()` places  $5 \times 5 = 25$  nodes, 600 m apart with a random offset of up to  $\pm 100$  m each way, over  $2.4 \text{ km} \times 2.4 \text{ km}$ . The centre point is the rescue base, so there are **24 relays + 1 base**.

In the seed-7 demo network, each relay has 2–6 neighbours within 800 m (mean 3.9), and the farthest relay is 4 hops from the base, inside the hop limit of 6.

### Radio model (from `network.py`)

- A node hears every live node within `range_m` (flat ground, no terrain).
- Two transmissions overlapping at a receiver destroy each other (collision).
- A node cannot receive while it is transmitting (half-duplex).
- Each successful reception can still be lost at random (`loss_probability`).

## Scenarios, as run by `python run.py all`

| Scenario       | What happens                                                                                                                                                 | Runs                                                       |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Rerouting demo | A critical SOS from the farthest node plus 4 random SOS; the relay in the middle of its route is then destroyed and a second SOS is sent from the same node. | seed 7                                                     |
| Failures       | Destroy 0%, 10%, 20%, 30%, 40%, 50% of relays at random, then every surviving node sends one SOS within 30 s. Run for 30 min.                                | 20 networks per row, seed 7                                |
| Load           | 10, 50, 100, 200, 500 SOS from random nodes within 1 min, no nodes destroyed. Run for 1 h. Priority mix: critical 30%, urgent 30%, supplies 20%, safe 20%.   | 5 networks per row ( <code>-- trials // 4</code> ), seed 7 |
| Health         | Every relay sends a heartbeat every 15 min; 3 nodes die silently after 1 h; the base checks at 3 h and flags nodes silent for over 30 min.                   | 10 runs ( <code>-- trials // 2</code> ), seed 7            |
| Sensitivity    | The failure test at ranges 500 m, 650 m, 800 m, 1000 m and random loss 0%, 2%, 5%, 10%, with 0%, 30% of relays destroyed.                                    | 20 networks per row, seed 7                                |

Defaults read from `scenarios.py` and `run.py` (`-- trials 20`, `-- seed 7`). The README reports that seeds 3 and 11 gave the same main conclusions.

## Why this layout (README section 4)

| Value                               | Reason                                                                                                                                                                                                       |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25 nodes, 2.4 km × 2.4 km           | Roughly neighbourhood-sized                                                                                                                                                                                  |
| 600 m spacing, ±100 m random offset | Below the 800 m range, so each node normally reaches its nearest neighbours but rarely its diagonal ones (about 850 m away). This forces multi-hop routes. The offset stops the grid being perfectly regular |
| Base at the centre                  | Farthest nodes are 4 hops away                                                                                                                                                                               |
| 2% random loss                      | Allowance for interference the model doesn't cover; Section 3.5 shows results hardly change from 0% to 10%                                                                                                   |

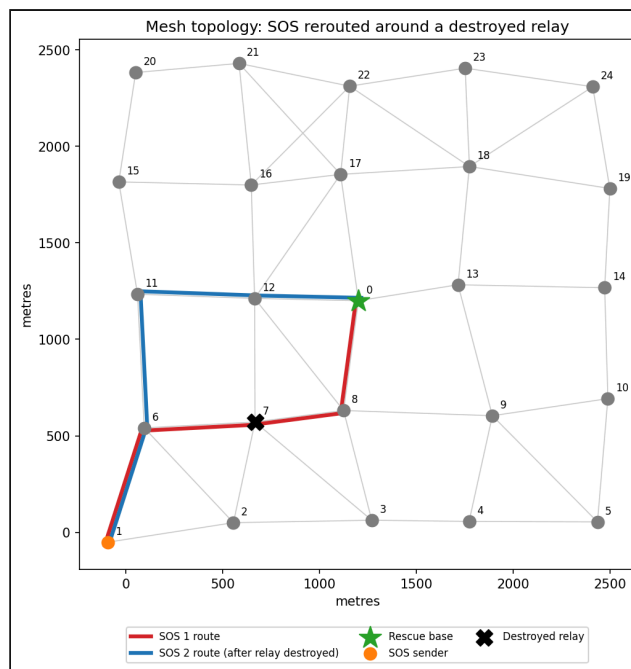
## 8 Results

Every CSV in `emergency-mesh/simulation/results/`, with every column. Column names are shown under each heading.

### Rerouting demo: rescue dashboard

| Priority<br>priority | Type<br>type | Node<br>node | Latitude<br>lat | Longitude<br>lon | People<br>people | Flags<br>flags       | Landmark<br>landmark | Hops<br>hops | Delay (s)<br>delay_s |
|----------------------|--------------|--------------|-----------------|------------------|------------------|----------------------|----------------------|--------------|----------------------|
| CRITICAL             | TRAPPED      | 1            | 12.98104        | 80.21713         | 4                | INJURED TRAPPED      | School rooftop       | 4            | 2.9                  |
| CRITICAL             | MEDICAL      | 11           | 12.99258        | 80.21855         | 7                | INJURED TRAPPED      | Water tank           | 2            | 1.5                  |
| URGENT               | FIRE         | 16           | 12.99767        | 80.22396         | 1                | CHILD                | Water tank           | 2            | 0.8                  |
| URGENT               | FLOODING     | 17           | 12.99816        | 80.22821         | 8                | CHILD                | Water tank           | 1            | 1.1                  |
| URGENT               | FLOODING     | 1            | 12.98104        | 80.21713         | 6                | ELDERLY WATER_RISING | School rooftop       | 4            | 1.9                  |
| SAFE                 | OTHER        | 18           | 12.99853        | 80.23434         | 5                | -                    | Water tank           | 2            | 6.5                  |

From `emergency-mesh/simulation/results/demo_dashboard.csv`; 6 rows, every column.

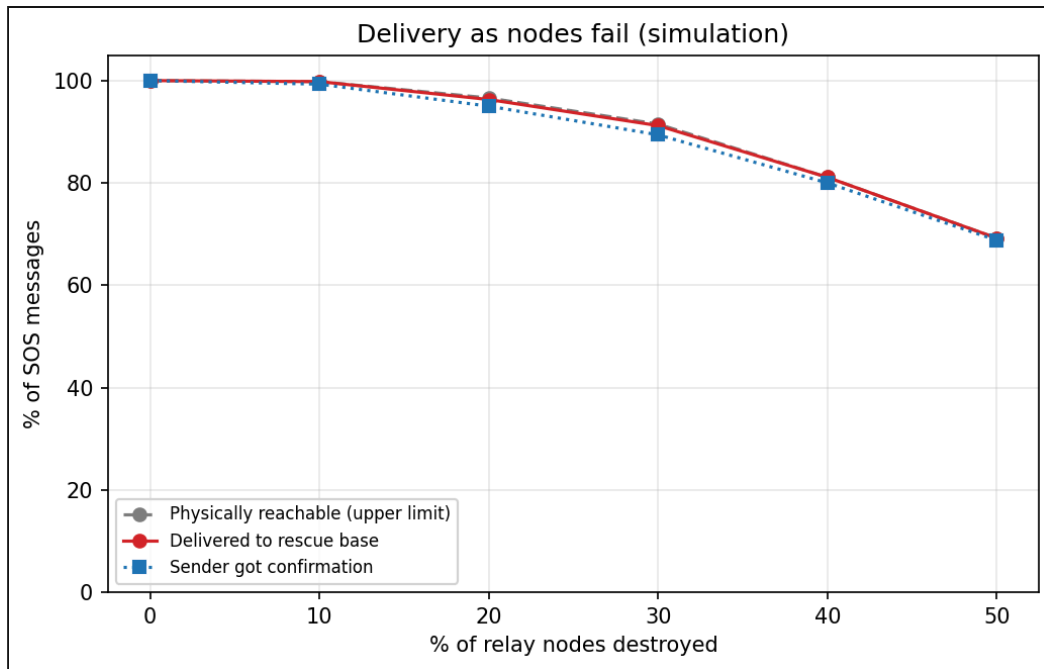


`results/demo_topology.png`: the demo network, both SOS routes and the destroyed relay

### Delivery as relays are destroyed

| Relays destroyed (%)<br>failed_pct | SOS sent<br>sos_snt | Reachable (%)<br>reachable_pct | Delivered (%)<br>delivered_pct | Delivered of reachable (%)<br>delivered_of_reachable_pct | Sender confirmed (%)<br>confirmed_pct | Median delay (s)<br>median_delay_s | 95th percentile delay (s)<br>p95_delay_s | Mean hops<br>mean_hops | Transmissions per SOS<br>tx_per_sos |
|------------------------------------|---------------------|--------------------------------|--------------------------------|----------------------------------------------------------|---------------------------------------|------------------------------------|------------------------------------------|------------------------|-------------------------------------|
| 0                                  | 480                 | 100.0                          | 100.0                          | 100.0                                                    | 100.0                                 | 12.5                               | 289.7                                    | 3.11                   | 45.2                                |
| 10                                 | 440                 | 99.8                           | 99.8                           | 100.0                                                    | 99.3                                  | 8.7                                | 291.0                                    | 3.03                   | 43.0                                |
| 20                                 | 380                 | 96.6                           | 96.3                           | 99.7                                                     | 95.0                                  | 6.9                                | 288.9                                    | 2.9                    | 35.8                                |
| 30                                 | 340                 | 91.5                           | 91.2                           | 99.7                                                     | 89.4                                  | 4.7                                | 288.3                                    | 2.67                   | 30.0                                |
| 40                                 | 280                 | 81.1                           | 81.1                           | 100.0                                                    | 80.0                                  | 3.5                                | 279.1                                    | 2.44                   | 21.3                                |
| 50                                 | 240                 | 69.2                           | 69.2                           | 100.0                                                    | 68.8                                  | 2.5                                | 280.1                                    | 2.39                   | 18.6                                |

From `emergency-mesh/simulation/results/failures.csv`; 6 rows, every column.

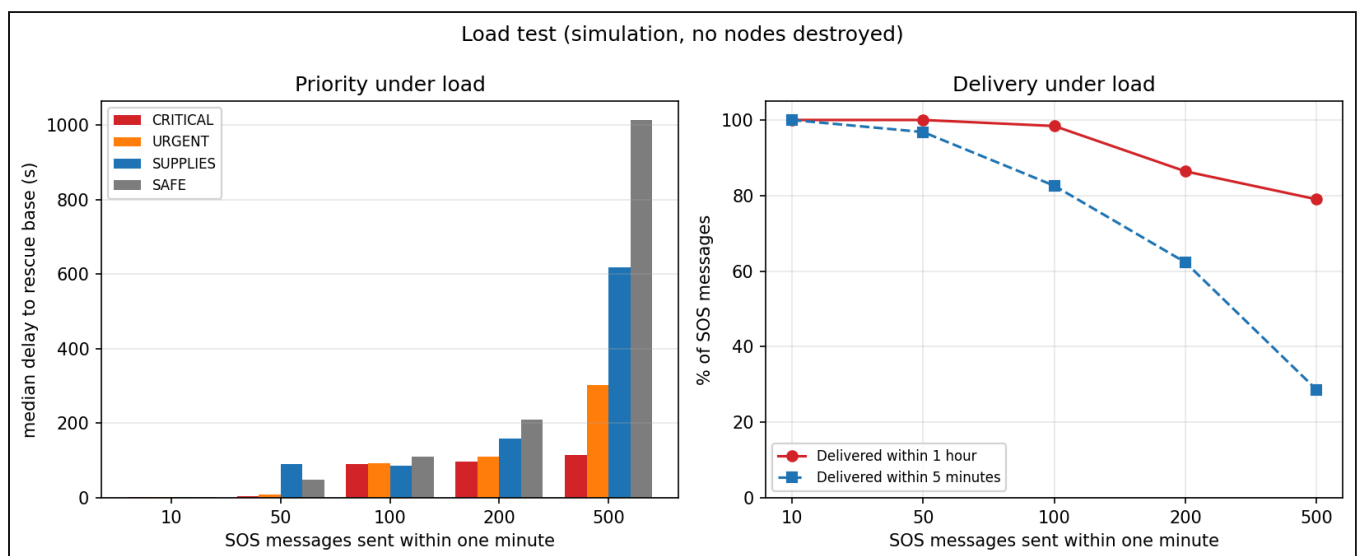


results/failures.png: delivery as relays are destroyed

## Many SOS at once

| SOS in 60 s<br>sos_in_60s | Delivered (%)<br>delivered_pct | Delivered within 5 min (%)<br>within_5min_pct | Median delay, critical (s)<br>median_delay_critical_s | Median delay, urgent (s)<br>median_delay_urgent_s | Median delay, supplies (s)<br>median_delay_supplies_s | Median delay, "I'm safe" (s)<br>median_delay_afe_s | 95th percentile delay (s)<br>p95_delay_s | Collisions per SOS<br>collisions_per_sos |
|---------------------------|--------------------------------|-----------------------------------------------|-------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------|----------------------------------------------------|------------------------------------------|------------------------------------------|
| 10                        | 100.0                          | 100.0                                         | 2.8                                                   | 2.4                                               | 1.9                                                   | 1.5                                                | 91.4                                     | 22.8                                     |
| 50                        | 100.0                          | 96.8                                          | 5.5                                                   | 8.8                                               | 91.1                                                  | 49.4                                               | 290.9                                    | 33.5                                     |
| 100                       | 98.4                           | 82.6                                          | 90.8                                                  | 92.7                                              | 85.8                                                  | 110.4                                              | 684.7                                    | 41.5                                     |
| 200                       | 86.4                           | 62.3                                          | 97.5                                                  | 110.5                                             | 159.9                                                 | 209.4                                              | 1387.2                                   | 40.3                                     |
| 500                       | 79.0                           | 28.6                                          | 114.4                                                 | 302.2                                             | 618.5                                                 | 1013.7                                             | 1429.0                                   | 50.3                                     |

From emergency-mesh/simulation/results/load.csv; 5 rows, every column.



results/load.png: delivery and delay under load

## Detecting dead nodes

| Runs<br>trials | Nodes killed<br>nodes_killed | Detected<br>detected | Missed<br>missed | False alarms<br>false_alarms | Mean time to alert (min)<br>mean_minutes_to_alert | Longest time to alert (min)<br>max_minutes_to_alert |
|----------------|------------------------------|----------------------|------------------|------------------------------|---------------------------------------------------|-----------------------------------------------------|
| 10             | 30                           | 30                   | 0                | 0                            | 20.8                                              | 28.0                                                |

From `emergency-mesh/simulation/results/health.csv`; 1 row, every column.

## Sensitivity to the radio assumptions

| Range<br>(m)<br>range_m | Random loss<br>(%)<br>loss_pct | Relays destroyed<br>(%)<br>failed_pct | Reachable<br>(%)<br>reachable_pct | Delivered<br>(%)<br>delivered_pct | Delivered of reachable<br>(%)<br>delivered_of_reachable_pct | Median delay<br>(s)<br>median_delay_s |
|-------------------------|--------------------------------|---------------------------------------|-----------------------------------|-----------------------------------|-------------------------------------------------------------|---------------------------------------|
| 500                     | 2                              | 0                                     | 0.0                               | 0.0                               | n/a                                                         | n/a                                   |
| 500                     | 2                              | 30                                    | 0.0                               | 0.0                               | n/a                                                         | n/a                                   |
| 650                     | 2                              | 0                                     | 82.7                              | 82.3                              | 99.5                                                        | 6.7                                   |
| 650                     | 2                              | 30                                    | 52.1                              | 52.1                              | 100.0                                                       | 2.7                                   |
| 800                     | 2                              | 0                                     | 100.0                             | 100.0                             | 100.0                                                       | 12.5                                  |
| 800                     | 2                              | 30                                    | 91.5                              | 91.2                              | 99.7                                                        | 4.7                                   |
| 1000                    | 2                              | 0                                     | 100.0                             | 100.0                             | 100.0                                                       | 13.6                                  |
| 1000                    | 2                              | 30                                    | 99.7                              | 99.7                              | 100.0                                                       | 6.2                                   |
| 800                     | 0                              | 0                                     | 100.0                             | 100.0                             | 100.0                                                       | 9.9                                   |
| 800                     | 0                              | 30                                    | 91.5                              | 91.5                              | 100.0                                                       | 4.8                                   |
| 800                     | 5                              | 0                                     | 100.0                             | 100.0                             | 100.0                                                       | 11.2                                  |
| 800                     | 5                              | 30                                    | 91.5                              | 91.5                              | 100.0                                                       | 6.1                                   |
| 800                     | 10                             | 0                                     | 100.0                             | 100.0                             | 100.0                                                       | 13.1                                  |
| 800                     | 10                             | 30                                    | 91.5                              | 91.2                              | 99.7                                                        | 5.3                                   |

From `emergency-mesh/simulation/results/sensitivity.csv`; 14 rows, every column.

## 9 Duty-cycle check (our estimate)

**Our own estimate, not a measurement.** Under India's rules for the 865–868 MHz band (README section 4), each relay may transmit at most 2.5% of the time. We assume a 1-hour observation period. Because every relay repeats every message, each relay's load is the whole network's traffic.

| Quantity                             | Calculation                                                 | Result                                       |
|--------------------------------------|-------------------------------------------------------------|----------------------------------------------|
| Heartbeat size and time on air       | 16 B at SF9                                                 | 164.9 ms                                     |
| Heartbeats per relay per hour        | $3600 \text{ s} / 900 \text{ s}$                            | 4                                            |
| Heartbeats each relay sends per hour | 24 relays $\times$ 4: its own, plus each other relay's once | 96                                           |
| Heartbeat time per relay per hour    | $96 \times 164.9 \text{ ms}$                                | $15.8 \text{ s} = \mathbf{0.44\%}$           |
| Budget at the limit                  | 2.5% of 3600 s                                              | 90 s                                         |
| Left for SOS traffic                 | $90 \text{ s} - 15.8 \text{ s}$                             | 74.2 s                                       |
| Time per SOS at each relay           | SOS 31 B (246.8 ms) + its ACK 13 B (164.9 ms)               | 411.6 ms                                     |
| SOS per hour before the limit        | $74.2 \text{ s} / 411.6 \text{ ms}$                         | <b>about 180</b> , across the whole network  |
| The 500-SOS load test                | $15.8 \text{ s} + 500 \times 411.6 \text{ ms}$              | $222 \text{ s} = \mathbf{6.2\%}$ of the hour |

- Assumes each relay forwards each SOS and its ACK exactly once (duplicates are dropped), with no retries. Retries in a real mass event only add to this.
- Heartbeats alone use 0.44% of each relay's time. The 500-SOS load test sends 500 SOS within 1 min, about 2.8 times what fits in a whole hour under the limit.
- Home units use ESP-NOW on 2.4 GHz, a different band, so they don't count against this limit.
- Gradient routing (README section 10) would cut how many relays repeat each message.

## 10 Outside facts

Facts this design relies on that come from outside the repository. None of them were measured by us. The list lives in `docs/tools/outside_facts.csv`, which has the full links. Each link was opened and checked against its fact on 10 October 2026.

| Fact                                                                                                                                                                                                                                                                                  | Source                                                                                                                                                                                                               | Link                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 865–868 MHz is licence-exempt for short range devices. For tracking, tracing and data acquisition devices: at most 500 mW e.r.p., channels up to 200 kHz, transmitting at most 2.5% of the time (10% for network access points), adaptive power control, and type-approved equipment. | DoT, Use of Low Power Equipment in the Frequency Band 865–868 MHz for Short Range Devices (Exemption from Licence) Rules, 2021 (G.S.R. 853(E)), Table II and rule 5; in DoT's compilation of subordinate legislation | <a href="https://eservices.dot.gov.in">eservices.dot.gov.in</a> (PDF, page 136) |
| The 2.4 GHz band used by ESP-NOW is licence-exempt in India.                                                                                                                                                                                                                          | DoT notification G.S.R. 45(E), 28 January 2005, rule 3; in DoT's compilation of subordinate legislation                                                                                                              | <a href="https://eservices.dot.gov.in">eservices.dot.gov.in</a> (PDF, page 143) |
| The SX1276's maximum transmit power is +20 dBm (100 mW). Time on air follows Semtech's published formula, used in <code>protocol.py</code> .                                                                                                                                          | Semtech SX1276 product page (links the datasheet)                                                                                                                                                                    | <a href="https://semtech.com">semtech.com</a>                                   |
| ESP-NOW carries up to 250 bytes per packet (v1) or 1470 bytes (v2). At most 20 paired peers, 17 of them encrypted. Broadcast and unencrypted packets are received without pairing. Encryption uses CCMP.                                                                              | Espressif ESP-IDF documentation, ESP-NOW                                                                                                                                                                             | <a href="https://docs.espressif.com">docs.espressif.com</a>                     |
| ESP-NOW with built-in antennas: about 150 m reliable, about 60% delivery at 300 m in an open field, and about 100% to 450 m in long-range mode. Not measured by us.                                                                                                                   | Espressif developer blog, "ESP-NOW for outdoor applications" (community authors, University of Žilina)                                                                                                               | <a href="https://developer.espressif.com">developer.espressif.com</a>           |
| HD44780 LCD controllers store only 8 custom characters.                                                                                                                                                                                                                               | Hitachi HD44780U datasheet, page 2 (SparkFun copy)                                                                                                                                                                   | <a href="https://cdn.sparkfun.com">cdn.sparkfun.com</a> (PDF)                   |
| 500 m–1 km range at SF9 in dense Indian cities (Delhi, Mumbai, Bangalore) for an SX1276 at +20 dBm with simple antennas. Our assumed 800 m range sits inside it.                                                                                                                      | Zbotic, SX1276 LoRa module range, sensitivity and spreading factor guide                                                                                                                                             | <a href="https://zbotic.in">zbotic.in</a>                                       |

## 11 Wokwi prototype pins

The node interface prototype (OLED screen, SOS button, RGB LED) runs in the Wokwi ESP32 simulator: <https://wokwi.com/projects/476841677745845249>. Pins from README section 6, checked against `emergency-mesh/wokwi_node/main.py`.

| Component                     | ESP32 pin         |
|-------------------------------|-------------------|
| OLED SSD1306, SCL / SDA       | GPIO 22 / 21      |
| SOS button (internal pull-up) | GPIO 14           |
| RGB LED, red / green / blue   | GPIO 27 / 32 / 33 |

In `main.py`: I2C SCL = GPIO 22, SDA = GPIO 21; button = GPIO 14; LED red / green / blue = GPIO 27 / 32 / 33.

**System**

- Every home needs a unit with power during an outage, and it must be within ESP-NOW reach of a relay.
- Unused equipment decays; heartbeats and drills reduce this but don't remove it.
- Mass events cause heavy collisions and long waits for low-priority messages (Section 3.3).
- Indian rules limit each relay to transmitting 2.5% of the time. Our forwarding has every relay repeat every message, so a mass event (like the 500-SOS test) would exceed that. Normal use, mostly heartbeats, uses about 0.4% of each relay's time (our own calculation, Section 3.3).
- The home unit has no login (nobody can sign in mid-disaster), so false alarms are possible; rate limits reduce this.
- It only helps if a control room or rescue team runs a base station.

**Simulation**

- Range is a simple circle on flat ground: no terrain, buildings or fading beyond random loss.
- Range and loss come from published figures and assumptions (Section 4), not our own measurements.
- Battery use is not modelled.
- Home units and ESP-NOW are not in the Python simulator; it models relays and the base only. The demo's 200 m home unit reach is an assumption.

**Not measured**

- Radio range (LoRa and ESP-NOW) and battery life have not been measured on real hardware.

From README section 9.